

Network Synthesis

→ so for we discuss N/W Analysis i.e

for given N/W { I response
voltage || are found.
Z funⁿ
Y funⁿ

but
* In the N/W synthesis it is reverse. i.e funⁿ is given, you have to develop the N/W.

Single port { $z(s), Y(s)$ } Impedance

Two port { 1. $z_{11}(s), z_{22}(s), Y_{11}(s), Y_{22}(s)$ } → Driving point Impedance function
2. $z_{12}(s), z_{21}(s), Y_{12}(s), Y_{21}(s)$ } → Transfer function
 $g_{12}(s), g_{21}(s), \alpha_{12}(s), \alpha_{21}(s)$

* In the N/W Analysis for the given N/W it is possible to find either current response (or) voltage response (or) impedance function (or) admittance function.

* In the N/W synthesis for a given funⁿ N/W is designed. (it is the reverse procedure of N/W Analysis)

* In the N/W synthesis it is possible to design the N/W for the following functions. (as shown above single port & two port).

→ In the N/W Analysis { Responses
Z-functions

we can use Theorems, Mesh Analysis, for all cases Answer is same, i.e unique solution is possible.

But we can design diff N/Ws. Here we don't get unique solution.

In the N/W synthesis it is possible to design 3 N/Ws.

- 1) Series N/W $\rightarrow$ obtained by Foster first form
 - 2) Parallel N/W $\rightarrow$ Foster second form
 - 3) Ladder N/W $\rightarrow$
 - Cauer - I form
 - Cauer - II form
- } partial fraction of expansion used.
- } Continued fraction of expansion.

* In the N/W Analysis always unique solution is obtained, but in the N/W synthesis for the give function it is possible to design the above three N/Ws.

Note: -) While designing the N/W from the given function, first of all it satisfy the necessary condⁿ $\rightarrow$ $f(s)$ should be proper & real funⁿ.

In the N/W analysis we concentrate on Active & Passive elements but here we concentrate only on passive elements & also NO -ve values of R/L/C are present i.e

always

$R \geq 0$
 $L \geq 0$
 $C \geq 0$

Ex: - $z(s) = -4 - 3s$
 $z = R + sL$
 $(R = -4, L = -3)$ so $z(s)$ is NOT a PRF.

1) If $f(s)$ is PRF (true real funⁿ) $\Rightarrow \frac{1}{f(s)}$ also PRF.

Ex: - $z(s)$ is a PRF $\Rightarrow Y(s)$ also PRF.

3) If f_1 & f_2 are PRF $\Rightarrow F(s) = f_1(s) + f_2(s)$ must be PRF.

Ex: - If z_1 & z_2 are true real funⁿ connected in series $\Rightarrow z_{eq}$ also PRF.

$\rightarrow$ All the poles of the function should be present in left half of the plane. i.e stability condition must be satisfy.

$\rightarrow$ Imaginary poles & zeros should be conjugate pair.

$\rightarrow$ In the partial fraction of expansion residue should be true real.

$\rightarrow$ Numerator & denominator polynomials should satisfy "Hurwitz"

$\rightarrow$ Highest powers of Numerator & denominator polynomials should be differ by atmost unity. This condition prohibits Multiple poles & zeros at infinite.

$\rightarrow$ Lowest powers of Numerator & denominator polynomials should be differ by atmost unity, this condition prohibits multiple poles & zeros at origin.

→ Total No. of poles in the function should be equal to Total No. of zeros.

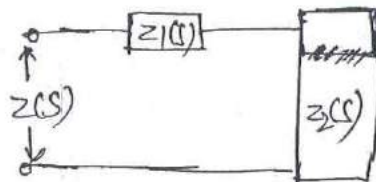
Note:- $Z(s)$ divided into two subfunⁿ.

$$Z(s) = Z_1(s) + Z_2(s)$$

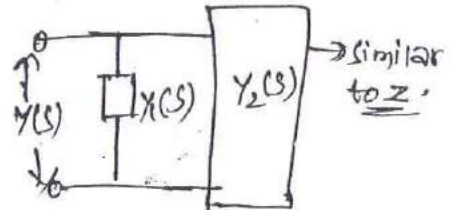
$$Z_2(s) = Z(s) - Z_1(s)$$

$$Y(s) = Y_1(s) + Y_2(s)$$

$$\Rightarrow Y_2(s) = Y(s) - Y_1(s)$$



again this is divided into two subfunⁿ like we this procedure is continued until total funⁿ is realized.



Note:- Based on removal of operation different procedures are developed.

→ Removal of Pole at ∞ :-

$$Z(s) = \frac{b_{n+1}s^{n+1} + b_n s^n + \dots + b_0}{a_n s^n + a_{n-1} s^{n-1} + \dots + a_0}$$

if $Z(s)$ pole is present at ∞ or not? since zero^s = $n+1$, pole^s = n . so one pole is at ∞ .

By long division

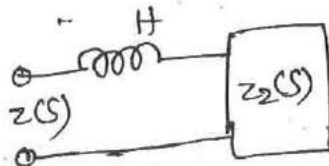
$$Z(s) = \frac{b_{n+1}s^{n+1}}{a_n s^n} + Z_2(s)$$

where $H = \frac{b_{n+1}}{a_n}$

$$Z(s) = \frac{H(s)^{n+1}}{s^n} + Z_2(s) \Rightarrow Z(s) = H \cdot s + Z_2(s)$$

$$\Rightarrow Z_2(s) = Z(s) - \frac{H(s)}{s} = Z(s) - H \cdot s$$

corresponds to $\frac{H(s)}{s}$'s' is present at numerator. so it is inductor.



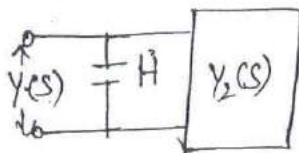
w.r.t $X = sL$ so $L = \frac{H}{s}$

Now for admittance funⁿ

$$Y(s) = (H(s)) + Y_2(s)$$

$$Y(s) = (H(s)) + Y_2(s) \Rightarrow Y_2(s) = Y(s) - H(s)$$

$$Y_2(s) = Y(s) - H \cdot s$$



2. Removal of pole at origin :-

$$Z(s) = \frac{b_0 + \dots + b_{n-1}s^{n-1} + b_n s^n}{s(a_0 + \dots + a_{n-1}s^{n-1} + a_n s^n)}$$

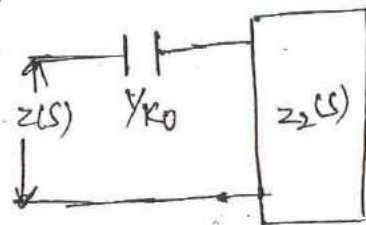
By long division...

correspond to $Z(s)$ pole is present at origin.

$$Z(s) = \frac{b_0}{s \cdot a_0} + Z_2(s) \quad \left[\text{where } \frac{b_0}{a_0} = k_0 \right]$$

$$Z(s) = \frac{k_0}{s} + Z_2(s) \Rightarrow Z_2(s) = Z(s) - \frac{k_0}{s} \dots$$

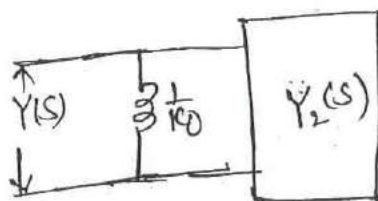
corresponding to impedance funⁿ 's' is present at denominator
so it is a capacitor.



The procedure, what we are doing for impedance funⁿ is same for admittance funⁿ.

$$Y(s) = \frac{k_0}{s} + Y_2(s) \dots$$

$$Y_2(s) = Y(s) - \frac{k_0}{s}$$



3. Removal of conjugate pair of poles :-

$$Z(s) = z_1(s) + z_2(s)$$

$$z_2(s) = Z(s) - z_1(s)$$

$$z_1(s) \rightarrow \text{poles} \rightarrow \pm j\omega$$

$$z_1(s) = \frac{k_1}{s + j\omega} + \frac{k_2}{s - j\omega}$$

Diw^{ays} k_1 & k_2 should be +ve. ($k_1 = k_2 = K$)

$$Z(s) = \frac{k}{s+j\omega} + \frac{k}{s-j\omega}$$

$$Z_1(s) = \frac{2ks}{s^2 + \omega^2} // \dots$$

$$Z_1(s) = \frac{1}{Y_a + Y_b}$$

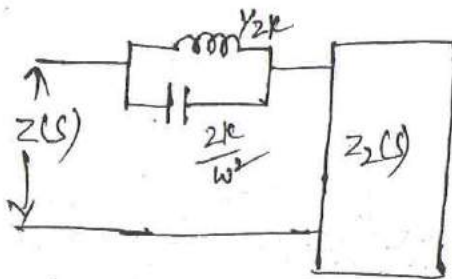
where

$$Y_a = sC = B_C$$

$$Y_b = \frac{1}{Ls} = B_L$$

That means when ever we are removing the $Z_1(s)$, we are removing parallel combination of LC .

$$\therefore Z(s) = Z_1(s) + Z_2(s)$$



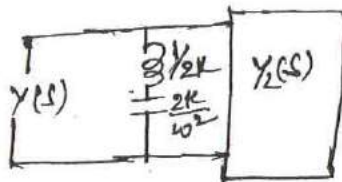
* The steps are same for admittance function.

$$Y(s) = Y_1(s) + Y_2(s)$$

$$Y_2(s) = Y(s) - Y_1(s)$$

$$Y_1(s) = \frac{1}{\left[\frac{s}{2k}\right] + \left[\frac{1}{\omega^2} \cdot \frac{1}{2ks}\right]} = \frac{1}{Z_a + Z_b}$$

$X_L = Ls$ $X_C = 1/sC$

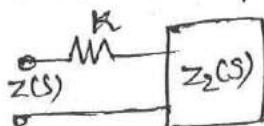


4. Removal of constant :-

$$Z(s) = k + Z_2(s)$$

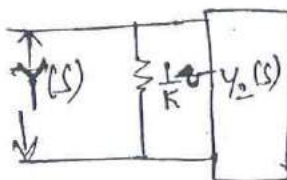
$$Z_2(s) = Z(s) - k$$

To obtain New funⁿ resistor k removed from $Z(s)$.



$$Y(s) = K + Y_2(s)$$

$$\Rightarrow Y_2(s) = Y(s) - K$$



(pb)

obtain Foster I & II form of $z(s)$?

$$z(s) = \frac{(s^2+2)(s^2+4)}{(s)(s^2+3)}$$

Sol:- Foster - I form :-

Numerator power > Denominator power. Hence removal of Pole at ∞ .

Pole at origin is present.
conjugate pair poles are present.

$$\frac{(s^2+2)(s^2+4)}{s(s^2+3)} = \frac{k_0}{s} + \frac{2ks}{s^2+3} + HS$$

$$\frac{s^4 + 6s^2 + 8}{s(s^2+3)} = \frac{A}{s} + \frac{Bs+C}{s^2+3}$$

$$\frac{(s^2+2)(s^2+4)}{s(s^2+3)} =$$

$$\frac{k_0(s^2+3) + 2ks(s) + HS(s(s^2+3))}{s(s^2+3)}$$

By comparing both sides

$$K_0 = 8/3$$

$$2K_1 \rightarrow \frac{1}{3}$$

$$H \rightarrow 1$$

$$Z(s) = \frac{8/3}{s} + \frac{s/3}{s^2+3} + s$$

$$= \frac{1}{\frac{3}{8}s} + \frac{1}{\frac{s^2+3}{\frac{s}{3} \cdot \frac{s}{3}}} + s$$

$$= \frac{1}{(\frac{3}{8}s)} + \frac{1}{3s + (\frac{9}{s})} + s$$

$B_c = s$ $B_d = \frac{1}{s}$

$\frac{1}{Y_a + \frac{1}{Y_b}}$

foaster-II form:-

$$Z(s) = \frac{(s^2+2)(s^2+4)}{s(s^2+3)}$$

(P)

Sol:-

Here we concentrate on Admittance funⁿ.

$$\text{so } Y(s) = \frac{s(s^2+3)}{(s^2+2)(s^2+4)}$$

Note:- * for example if the given pblm $F(s) = \frac{(s^2+3)(s^2+4)}{s(s^2+3)}$ is given same funⁿ is utilized for foaster I & foaster-II form.

* if $Z(s)$ is given to obtain the foaster second form $\frac{1}{Z(s)}$ function is used.

* if $Y(s)$ is given to obtain the foaster-I form $\frac{1}{Y(s)}$ function is used.

Since numerator power < denominator,

$\Rightarrow$ Poles at ∞ is not present.

$\rightarrow$ pole at origin is not present

$\rightarrow$ conjugate pair of poles are present.

$$Y(s) = \frac{2K_1s}{s^2+\omega_1^2} + \frac{2K_2s}{s^2+\omega_2^2}$$

$$\frac{s(s^2+3)}{(s^2+2)(s^2+4)} = \frac{2K_1s}{s^2+\omega_1^2} + \frac{2K_2s}{s^2+\omega_2^2}$$

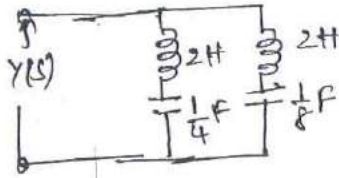
$$s(s^2+3) = 2K_1s(s^2+\omega_1^2) + 2K_2s(s^2+\omega_2^2)$$

where $\omega_1^2 = 2$
 $\omega_2^2 = 4$

$$\begin{aligned} \Rightarrow 2K_1 &= \frac{1}{2} \\ 2K_2 &= \frac{1}{2} \\ \dots \end{aligned}$$

$$\therefore Y(s) = \frac{s/2}{s^2+2} + \frac{s/2}{s^2+4}$$

$$= \frac{1}{\frac{s^2}{s/2} + \frac{2}{s/2}} + \frac{1}{\frac{s^2}{s/2} + \frac{4}{s/2}}$$



$$Y(s) = \frac{1}{2s + \frac{4}{s}} + \frac{1}{2s + \frac{8}{s}}$$

$\swarrow$ $\searrow$
 $z_a + z_b$ $z_c + z_d$
 $x_c = \frac{1}{s}$

LC Network :-

Cauer-I form :-

Removable of pole at ∞ :-

corresponding to impedance funⁿ (as shown in dgm) a pole at ∞ is present. Now the new funⁿ which is result of removal of pole at ∞ .

$$z_2(s) = z(s) - H_1 s \Rightarrow x_L = Ls$$

$$Y_2(s) = \frac{1}{z_2(s)} \rightarrow \text{(Pole is replaced zero \& vice versa)}$$

Pole at ∞ is removed; then resultant new funⁿ is

$$\therefore Y_3(s) = Y_2(s) - (H_2)s \Rightarrow x_C = \frac{1}{s}$$

again reciprocal of $Y_3(s) = z_3(s)$

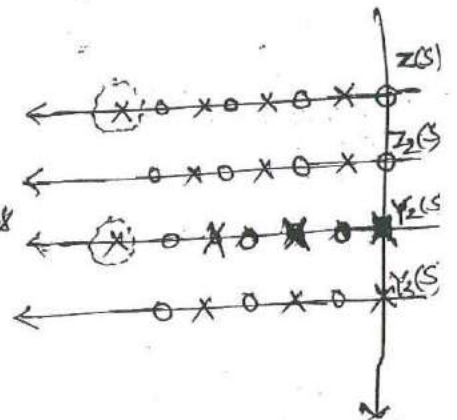
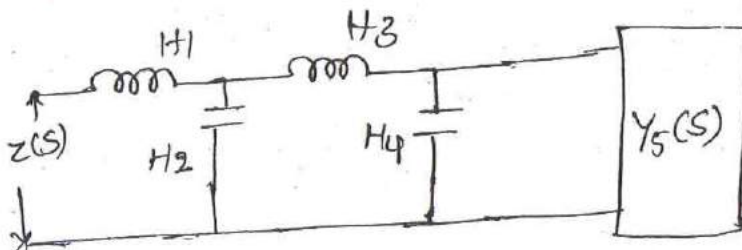
$$\therefore z_3(s) = \frac{1}{Y_3(s)}$$

$$z_4(s) = z_3(s) - (H_3)s \Rightarrow x_L = Ls$$

$$\text{again } Y_4(s) = \frac{1}{z_4(s)}$$

$$Y_5(s) = Y_4(s) - H_4 s \Rightarrow x_C = \frac{1}{s}$$

By rearranging the above funⁿ $\Rightarrow Y_4(s) = Y_5(s) + H_4 s$
 This procedure is repeated until total function is realized.



LC Network:

Cauer - II form:-

Removal of pole at origin:-

$$z_2(s) = z(s) - \frac{k_{01}}{s} \quad x_c = 1/s$$

$$y_2(s) = \frac{1}{z_2(s)}$$

$$y_3(s) = y_2(s) - \frac{k_{02}}{s} \quad B_L = 1/s$$

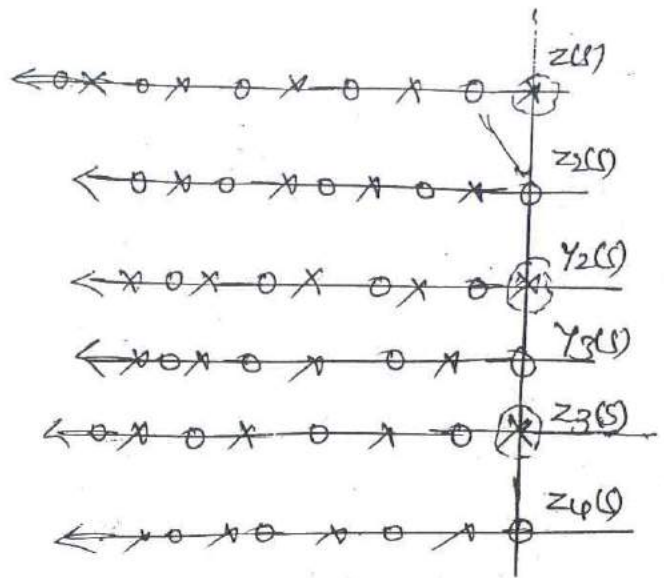
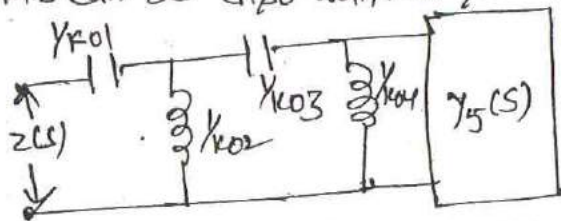
$$z_3(s) = \frac{1}{y_3(s)}$$

$$z_4(s) = z_3(s) - \frac{k_{03}}{s} \quad x_c = 1/s$$

$$y_4(s) = \frac{1}{z_4(s)}$$

$$y_5(s) = y_4(s) - \frac{k_{04}}{s} \quad B_L = 1/s$$

This can be also written as $y_4(s) = y_5(s) + \frac{k_{04}}{s}$



(Pb) obtain Cauer I & II form of $z(s)$.

(Pb) $z(s) = \frac{s^3 + 2s}{s^4 + 5s^2 + 4}$

(Pb) sol:- In the Cauer - I & II form continued fraction of expansion is utilized.

Here pole at ∞ .

Cauer - I form:-

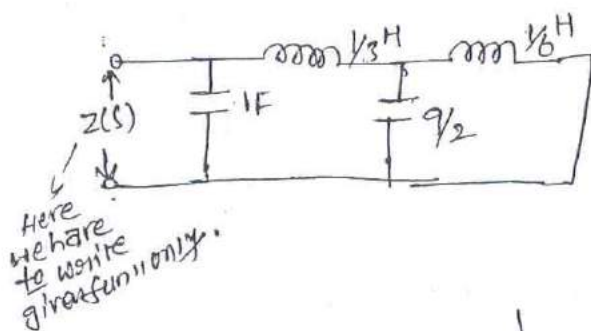
Removal of pole at ∞ .

$$y(s) = \frac{s^4 + 5s^2 + 4}{s^3 + 2s}$$

We are starting with $y(s)$. so first term $\rightarrow y$.

In this form alternately y & z forms are taken as shown in the Cauer I form derivation.

$$\begin{aligned} & \frac{s^3 + 2s}{s^4 + 5s^2 + 4} \xrightarrow{R_c = s} y \\ & \frac{s^4 + 5s^2}{s^3 + 2s} \xrightarrow{K_L = s} z \\ & \frac{5s^2 + 4}{s^3 + 2s} \xrightarrow{R_c = s} y \\ & \frac{5s^2}{s^3 + 2s} \xrightarrow{K_L = s} z \\ & \frac{5s^2}{s^3 + 2s} \xrightarrow{R_c = s} y \\ & \frac{5s^2}{s^3 + 2s} \xrightarrow{K_L = s} z \end{aligned}$$



Cauer-II form:-

$$Z(s) = \frac{s(s^2+2)}{s^4+s^2+4}$$

given impedance funn is not having pole at origin. so

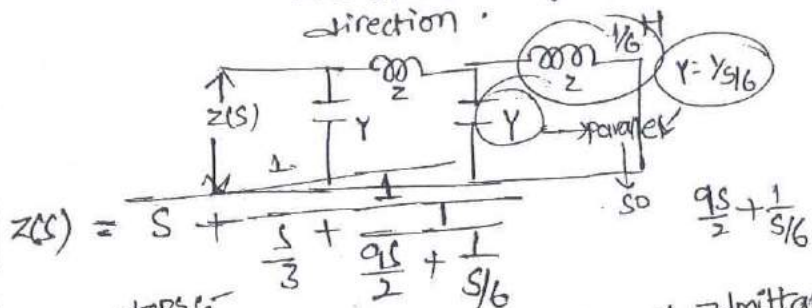
$$Y(s) = \frac{1}{Z(s)} = \frac{s^4+s^2+4}{s(s^2+2)}$$

Cauer I form :

we do expansion from highest power to lowest power, since our aim is to remove the poles at ∞ .

But in Cauer-II form the procedure is reverse, since our aim is to remove the zero poles at origin.

Note :- In some points they given the N/w. ^{asking} find out the funn. In that cases we are starting from backward direction.



steps:-

(i) $\frac{qs}{2}$ & $1/s/6$ are in the form of admittance
so $Y_{eq} = \frac{qs}{2} + 1/s/6$

(ii) Next inductor is connected. $Z_{eq} = \frac{1}{Y_{eq}}$
so both are in series.

(iii) Next parallel form (Y) is connected. so
 $Y_{eq} = 1/Z_{eq}$

$$s + \frac{1}{\frac{s}{3} + \frac{1}{\frac{qs}{2} + \frac{1}{s/6}}}$$

(iv) This is Y_{eq} resultant. but we need
 $Z_{eq} = \Rightarrow Z = \frac{1}{s + \frac{1}{\frac{s}{3} + \frac{1}{\frac{qs}{2} + \frac{1}{s/6}}}}$

$$\frac{2s^2+3}{s^4+s^2+4}$$

$$\frac{2s^2+3}{s^4+s^2+4} \left(\frac{2}{s} \right) \rightarrow Y \quad \rightarrow B_L = \frac{1}{s}$$

$$\frac{3s^2+4}{3s^2+s^4} \left(\frac{3}{3s} \right) \rightarrow Z \quad \rightarrow X_C = \frac{1}{3s}$$

$$\frac{s^3}{3s^2+s^4} \left(\frac{9}{s} \right) \rightarrow Y \quad \rightarrow B_L = \frac{1}{s}$$

$$\frac{s^4}{s^3} \left(\frac{1}{3s} \right) \rightarrow Z \quad \rightarrow X_C = \frac{1}{3s}$$

$$\frac{s^3}{3} \rightarrow 0$$

$$Z_{RC} = \frac{R \cdot \frac{1}{Cs}}{R + \frac{1}{Cs}}$$

$$= \frac{R}{RCs + 1} \quad // \dots = \frac{1}{C(s + \frac{1}{RC})}$$



$$Y_{RC} = \frac{1}{R + \frac{1}{Cs}}$$

$$= \frac{Cs}{RCs + 1}$$

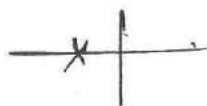
$$= \frac{s}{R(s + \frac{1}{RC})}$$



Pole-zero plot of both is not same.

Now by doing some modifications as shown in below, we get

$$\Rightarrow Z_{RC} = \frac{1}{C(s + \frac{1}{RC})}$$



$$\Rightarrow \frac{Y_{RC}}{s} = \frac{1}{R(s + \frac{1}{RC})} \quad //$$



$\therefore$ pole-zero plot is same.

Note:-

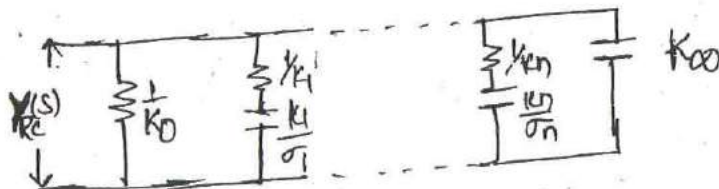
When ever pole zero plot of Z & Y are same, then for Y we use Z function, since both are same mathematical function can be use.

$$\frac{Y_{RC}}{s} = \left\{ \frac{K_0}{s} + \sum_{i=1}^n \frac{\frac{K_i}{s + \sigma_i}}{s + \sigma_i} + K_{\infty} \right\} \rightarrow Z(s) \text{ function}$$

$$Y_{RC} = \left(\frac{K_0}{s} + \sum_{i=1}^n \frac{K_i s}{s + \sigma_i} + K_{\infty} s \right)$$

$$\frac{KS}{s + \sigma_i} \Rightarrow \frac{\frac{1}{K}}{1 + \frac{\sigma_i}{KS}} = \frac{\frac{1}{K}}{1 + \frac{\sigma_i}{KS}}$$

$\downarrow$ $\downarrow$
 Z_a $Z_b = X_c = \frac{1}{sC}$
 $P = \frac{1}{K}$



Note:-

Z_{RC} & $\frac{Y_{RC}}{s}$ pole-zero patterns are identical. There by for both functions same mathematical function can be used.

Properties of LNW:

$$Z_L(s) = \frac{K_0}{s} + \sum_{i=1}^n \frac{2K_i s}{s^2 + \omega_i^2} + HS$$

obj. question

- *1. either pole (or) zero should be present at origin.
- *2. either pole (or) zero should be present at infinite.
- 3. poles and zeros are arranged alternately along jw axis.
- 4. an partial fraction of expansion residue should be positive real.

NOTE:-

$$Z = j(X_L - X_C)$$

i.e. for LNW impedance is nothing but a net reactance.
so Z is replaced by jX & s is replaced by $j\omega$.

$$jX = \frac{K_0}{j\omega} + \sum_{i=1}^n \frac{2K_i (j\omega)}{(j\omega)^2 + \omega_i^2} + (H)(j\omega)$$

$$\Rightarrow X = -\frac{K_0}{\omega} + \sum_{i=1}^n \frac{2K_i \omega}{-\omega^2 + \omega_i^2} + H\omega$$

now

$$\frac{dX}{d\omega} = +\frac{K_0}{\omega^2} + \frac{d(\dots)}{d\omega} + \frac{d(H\omega)}{d\omega}$$

$\downarrow$ $\downarrow$ $\downarrow$
 +ve +ve +ve

So $\frac{dX}{d\omega} \rightarrow +ve$ i.e. $\frac{dX}{d\omega} > 0$...

so slope is +ve.

5. $\frac{dX}{d\omega} > 0 \rightarrow \text{slope is +ve}$

Properties of $Z_{RC}(s)$ & $Y_{RL}(s)$:-

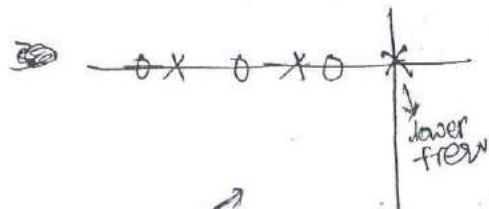
The pole zero plot of Z_{RC} & Y_{RL} both are same.

$$Z_{RC}(s) = \frac{K_0}{s} + \sum_{i=1}^n \frac{K_i}{s + \sigma_i} + K_\infty$$

pole at origin i.e. only lowest one.

1. lowest critical freq (1st critical freq) is due to pole, it may present either at origin (or) near to origin.

2. Poles & zeros are arranged alternately along -ve real axis.



always total no. of poles = total no. of zeros
so that for any poles & zeros ending is zero...

3. Highest critical frequency is due to zero, it may present either at infinite (or) nearer to infinite.

4. $\left. \begin{aligned} \frac{dZ_{RC}}{ds} < 0 \\ \frac{dY_{RL}}{ds} < 0 \end{aligned} \right\} \text{slope is } \underline{-ve}.$

$\left[\begin{aligned} Z_{RC} &= \frac{K_0}{s} + \dots \\ \frac{dZ_{RC}}{ds} &= -\frac{K_0}{s^2} + \dots \end{aligned} \right]$
-ve

$s=0$	$s=\infty$
$X_c = \frac{1}{sC} = \infty$	$X_c = \frac{1}{sC} = 0$

5. $\begin{aligned} Z_{RC}(0) &> Z_{RC}(\infty) \\ Y_{RL}(0) &> Y_{RL}(\infty) \end{aligned}$

simply says
there follows Z_{RC}
so $Y_{RL}(0) > Y_{RL}(\infty)$

Properties of $Z_{RL}(s)$ & $Y_{RC}(s)$:-

Pole-zero pattern of both are same, so same mathematical expⁿ can be used.

$$Y_{RC}(s) = K_0 + \sum_{i=1}^n \frac{-K_i s_i}{s + \sigma_i} + K_{\infty} s$$

1. Lowest critical frequency (First critical frequency) is due to zero, it may present either at origin (or) nearer to origin.
2. Poles & zeros are arranged alternately along -ve real axis.
3. Poles & zeros are alternate. Always No. of poles is equal to No. of zeros.

Highest critical frequency is due to pole, it may present either at infinite (or) nearer to infinite.

4. $Z_{RC}(s) \rightarrow$ previously discussed, ^{but here} it is opposite to $Z_{RC}(s)$ so slope is +ve instead of -ve.

4. $\left. \begin{aligned} \frac{dZ_{RL}(s)}{ds} > 0 \\ \frac{dY_{RC}(s)}{ds} > 0 \end{aligned} \right\} \text{slope is } \underline{+ve}.$

$S=0$	$S=\infty$
$X_L = j\omega L$ $=0$	$X_L = j\omega L$ $=\infty$

$$5. \boxed{Z_{RL}(0) < Z_{RL}(\infty)}$$

$$\boxed{Y_{RC}(0) < Y_{RC}(\infty)}$$

RC N/W:-

Cauer-I form:-

refn is taken as LC N/W. In the LC N/W we concentrate on the removal of pole at ∞ and then admittance function & then repeated this procedure.

$$LC \rightarrow Z_2(S) = Z(S) - H_1 S \quad ; \quad Y_2(S) = \frac{1}{Z_2(S)}$$

$$Y_3(S) = Y_2(S) - H_2 S$$

$\vdots$

RC $\rightarrow$ Step(1):- Removal of constant from $Z(S) \rightarrow$ (instead of removing the inductor we remove the resistor).

$$Z_2(S) = Z(S) - K_1 \text{ resistor}$$

$$Y_2(S) = \frac{1}{Z_2(S)}$$

Step(2):- Removal of pole at ∞ from $Y(S)$.

$$Y_3(S) = Y_2(S) - H_1 S \text{ capacitor } B_C(S) = CS$$

Cauer form $\rightarrow$ means always ladder N/W is obtained. that is after removing capacitor again we are removing the resistor.

$$Z_3(S) = \frac{1}{Y_3(S)}$$

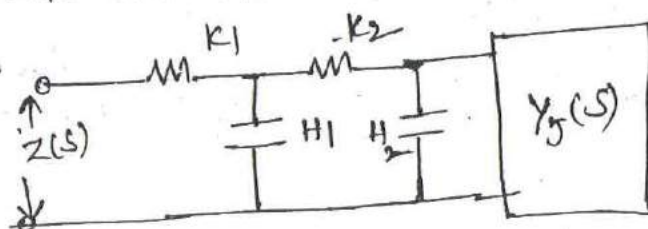
$$Z_4(S) = Z_3(S) - K_2 \text{ constant}$$

$$Y_4(S) = \frac{1}{Z_4(S)}$$

$$Y_5(S) = Y_4(S) - H_2 S \text{ } B_C(S) = CS$$

$$\Rightarrow Y_4(S) = Y_5(S) + H_2 S$$

Step(1) & Step(2) are repeated until the total funn is realized. based on this mathematical funn we can realize the N/W.



That means for LC N/W & RC N/W 'C' is same, 'L' is replaced by 'R'.

NOTE:- step 1 & step 2 are alternately repeated until the total funⁿ is realized.

RC N/W:-

Cauer-II form:-

Removal of pole at origin.

LC $\rightarrow z_2(s) = z(s) - \frac{k_{01}}{s}$, $y_2(s) = \frac{1}{z_2(s)}$, $y_3(s) = y_2(s) - \frac{k_{02}}{s}$

RC $\Rightarrow$ step(1):- Removal of pole from $z(s)$.

$$z_2(s) = z(s) - \frac{k_{01}}{s}$$

$$y_2(s) = \frac{1}{z_2(s)}$$

step(2):-

Removal of constant from $y(s)$

$$y_3(s) = y_2(s) - k_1$$

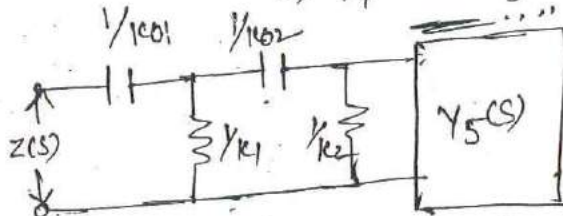
$$z_3(s) = \frac{1}{y_3(s)}$$

$$z_4(s) = z_3(s) - \frac{k_{02}}{s}$$

$$y_4(s) = \frac{1}{z_4(s)}$$

$$y_5(s) = y_4(s) - k_2$$

$$\Rightarrow y_4(s) = \frac{y(s)}{s} + k_2$$



Note:- in both RC N/W Cauer-I form & RC N/W Cauer-II form the elements are interchanged.

$\rightarrow$ capacitor & resistors are alternately repeated so this is called as "ladder N/W".

Note:- step(1) & step(2) are alternately repeated until the total funⁿ is realized.

(Pb)

Obtain Foster I & II form of $z(s) = \frac{(s+1)(s+4)}{s(s+2)}$

sol:-

$$z(s) = \frac{k_0}{s} + \dots$$

partial frac:

$$z(s) = \frac{A}{s} + \frac{B}{s+2} + C$$

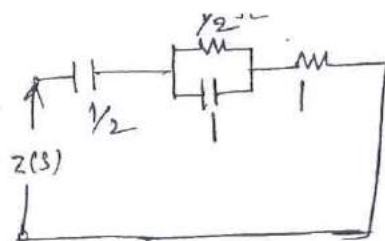
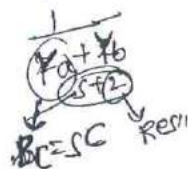
$$\Rightarrow (s+1)(s+4) = A(s+2) + Bs + C(s(s+2))$$

$$\Rightarrow \begin{cases} A = 2 \\ B = 1 \\ C = 1 \end{cases}$$

Note:- whenever numerator = denominator power additional constant term is present.

$$Z(s) = \left(\frac{2}{s} \right) + \left(\frac{1}{s+2} \right) + 1$$

$$X_C = \frac{1}{sC}$$



Foster-I form

Foster-II form :-

$$Y(s) = \frac{s(s+2)}{(s+1)(s+4)} = \frac{A}{s+1} + \frac{B}{s+4} + C$$

$$s^2 + 2s = As + 4A + Bs + B + Cs^2 + 4C + 5Cs$$

$$\Rightarrow A = -\frac{1}{3}$$

$$B =$$

$$C =$$

* In the partial fraction of expansion all the residues must be +ve.

✗ In the partial fraction expansion if -ve residue occurs then go for ~~Y(s)/s~~.

$$\frac{Y(s)}{s}$$

NOW

$$\frac{Y(s)}{s} = \frac{s(s+2)}{s(s+1)(s+4)}$$

$$\frac{s+2}{(s+1)(s+4)} = \frac{A}{s+1} + \frac{B}{s+4}$$

$$\Rightarrow A = \frac{1}{3}$$

$$B = \frac{2}{3}$$

$$\frac{Y(s)}{s} = \frac{\frac{1}{3}}{s+1} + \frac{\frac{2}{3}}{s+4}$$

$$\Rightarrow Y(s) = \frac{\frac{s}{3}}{s+1} + \frac{\frac{2s}{3}}{s+4}$$

$$= \frac{1}{3 + \frac{3}{s}} + \frac{2}{3 + \frac{3}{s}}$$

$$= \frac{1}{\left(3 + \frac{3}{s} \right)} + \frac{1}{\left(\frac{3}{2} + \frac{6}{s} \right)}$$

$$\frac{1}{2a + z_b} \dots \frac{1}{2a + z_b}$$

